

Week 6 – Building a Baseline Model

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GitHub Repository: [dontayerosemain7-git/carisurg-portfolio](https://github.com/dontayerosemain7-git/carisurg-portfolio)

Objective

The objective of this project was to develop and evaluate baseline machine learning models for predicting Emergency Severity Index (ESI) triage levels using the Mercer emergency department dataset. Two baseline classifiers, Logistic Regression and Decision Tree, were developed and compared with a stratified random baseline. The purpose was to establish a reproducible benchmark before considering more advanced machine learning approaches.

Results

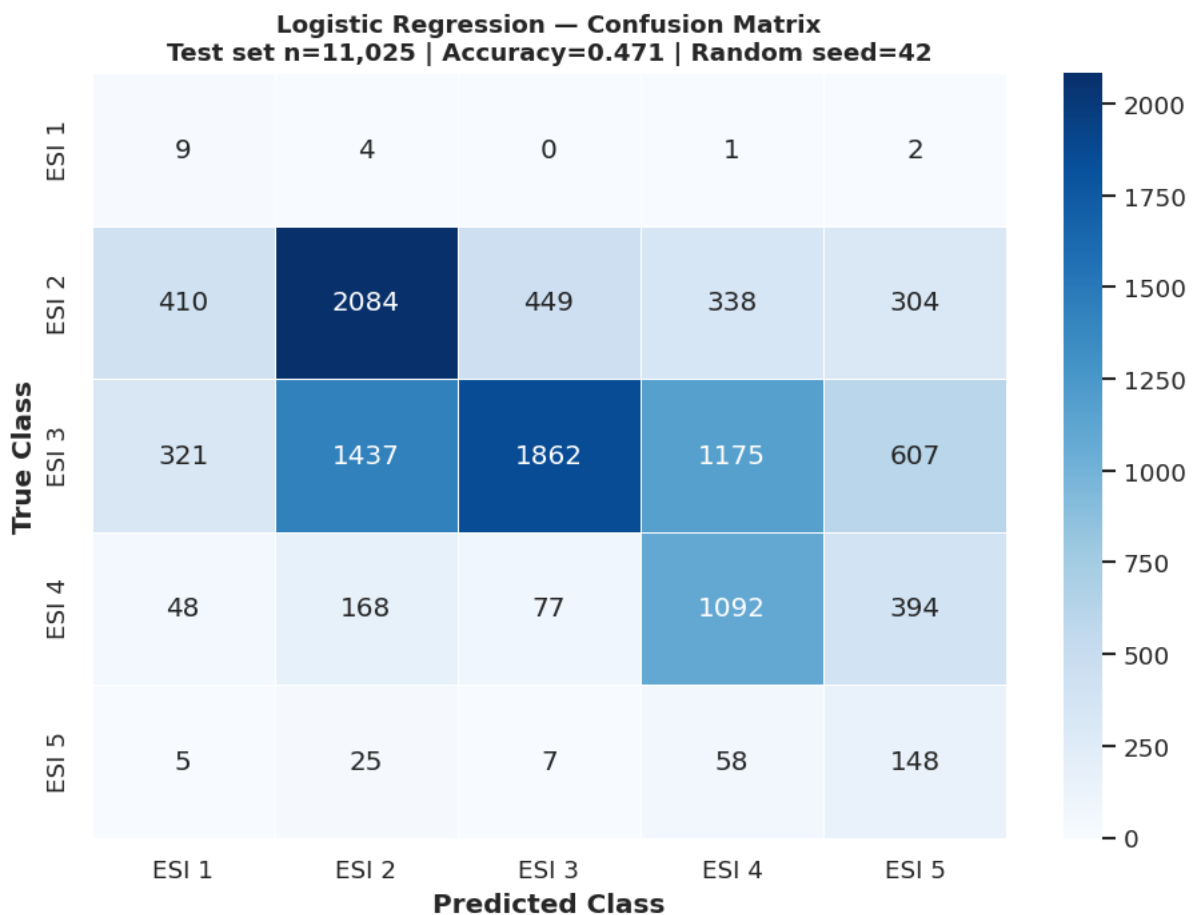


Figure 1. Confusion matrix for the Logistic Regression baseline model.

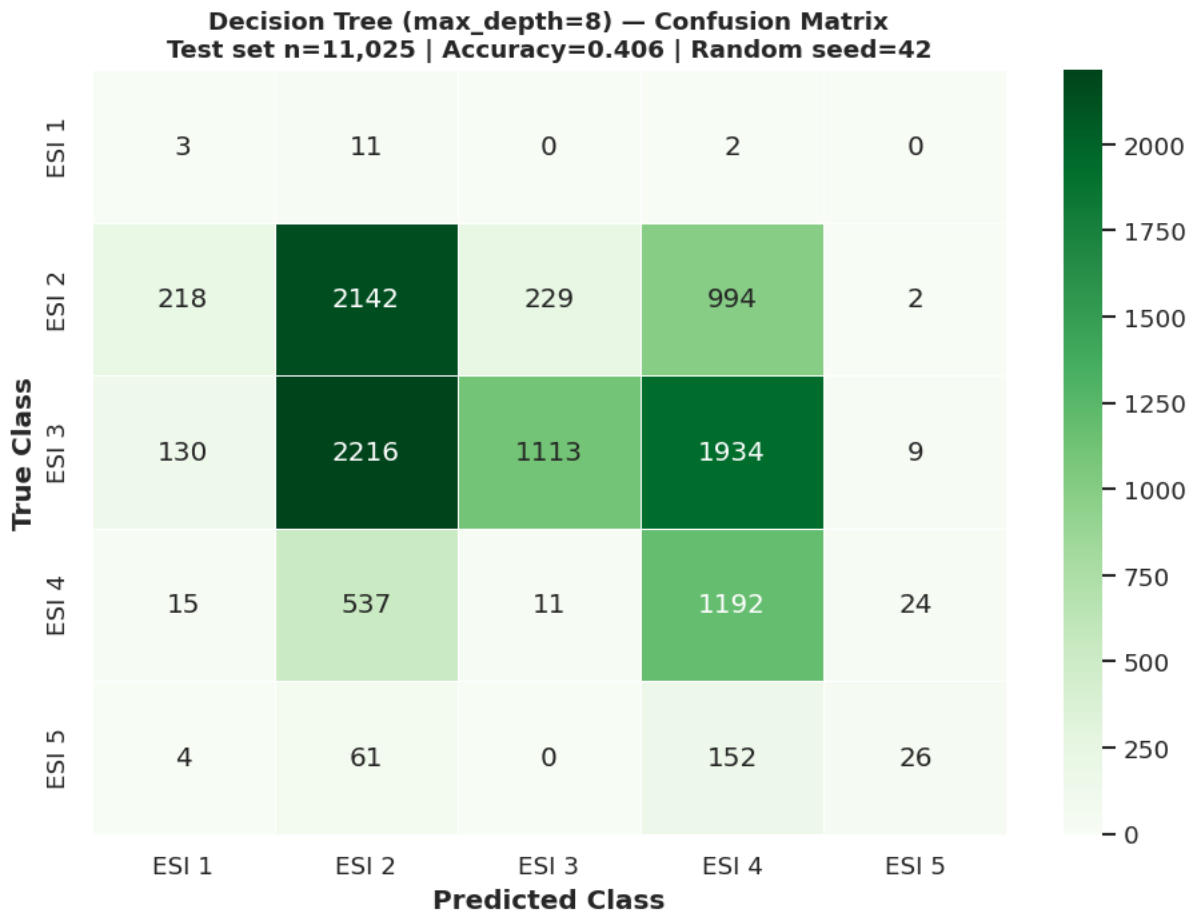


Figure 2. Confusion matrix for the Decision Tree baseline model.

The Logistic Regression model achieved the strongest overall performance of the three baseline models, with a test accuracy of **47.1%**, representing an improvement of **9.6 percentage points** over the stratified random baseline. In comparison, the Decision Tree achieved a test accuracy of **40.6%**, improving on the baseline by **3.1 percentage points**. The Logistic Regression model also produced a higher weighted F1-score (**0.5028** versus **0.3917**), indicating better overall performance across the class distribution. However, both models showed reduced performance on the less common triage categories, reflecting the challenges posed by class imbalance. Although the Logistic Regression model identified a greater proportion of **ESI Level 1** patients than the Decision Tree (**56.3%** compared with **18.8%**), performance for this highest-acuity class remained limited because only **16** ESI Level 1 cases were present in the test set. Overall, the Logistic Regression model was selected as the preferred baseline because it consistently outperformed both the Decision Tree and the stratified random baseline while providing a stronger foundation for future model development.

Primary Metric Justification

The primary evaluation metric selected for this project is **recall for ESI Level 1**. Patients assigned to ESI Level 1 require immediate life-saving intervention, and failing to identify these patients could delay critical treatment. Although overall accuracy provides a useful summary of performance, it can be misleading in an imbalanced dataset because the majority of patients

belong to lower-acuity triage categories. Prioritising recall for ESI Level 1 better reflects the clinical objective of minimising under-triage of critically ill patients, where the consequences of a missed prediction are substantially more serious than an unnecessary high-priority assessment.

Failure-Mode Reflection

The most concerning failure mode is the misclassification of ESI Level 1 patients into lower-acuity categories. Such errors could delay immediate assessment and treatment for patients experiencing life-threatening conditions such as cardiac arrest, septic shock, or major trauma. Although over-triage may increase emergency department workload, under-triage presents a greater risk to patient safety because critically ill patients may not receive timely intervention.

Conclusion

This project established two reproducible baseline models for predicting Emergency Severity Index classifications. Logistic Regression produced the strongest overall performance and provides an appropriate benchmark for future work. More advanced methods, including ensemble learning, feature engineering, and class-balancing techniques, may improve predictive performance; however, future models should continue to prioritise the identification of critically ill patients while maintaining clinical interpretability.